QUADRATICS, BINET AND PELL ON ONE PAGE

Metallic ratios: quadratics to Pell

Golden, silver and bronze: one quadratic connects rectangles, exact sequence formulas and integer equations.

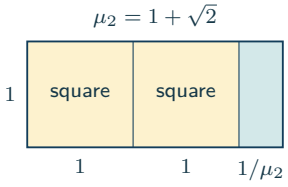
1 / REMOVE SQUARES AND RECOVER THE SAME SHAPE

One integer parameter gives the whole family

Remove λ unit squares from a rectangle with sides μ and 1, where $\lambda = 1, 2, 3, \ldots$. Require the remaining rectangle, turned through $90°$, to have the same aspect ratio. Then $\mu - \lambda = 1/\mu$. The λ-**metallic ratio** is the positive root:

$$\boxed{\mu^2 - \lambda\mu - 1 = 0, \qquad \mu_\lambda = \frac{\lambda + \sqrt{\lambda^2 + 4}}{2}}.$$

The other root is $\nu_\lambda = (\lambda - \sqrt{\lambda^2 + 4})/2 = -1/\mu_\lambda$.



Silver: remove two unit squares.

2 / WEIGHT THE LATEST TERM, THEN ADD THE PREVIOUS ONE

The same quadratic controls a sequence

For fixed $\lambda \geq 1$, set $U_0 = 0$, $U_1 = 1$ and $U_{k+2} = \lambda U_{k+1} + U_k$. Then $U_{k+1}/U_k \to \mu_\lambda$ as $k \to \infty$.

λ	ratio / sequence	quadratic	$U_0, \ldots, U_5$	positive root	value
1	Golden / Fibonacci	$x^2 - x - 1 = 0$	$0, 1, 1, 2, 3, 5$	$(1 + \sqrt{5})/2$	1.618034
2	Silver / Pell	$x^2 - 2x - 1 = 0$	$0, 1, 2, 5, 12, 29$	$1 + \sqrt{2}$	2.414214
3	Bronze	$x^2 - 3x - 1 = 0$	$0, 1, 3, 10, 33, 109$	$(3 + \sqrt{13})/2$	3.302776
4	4-metallic	$x^2 - 4x - 1 = 0$	$0, 1, 4, 17, 72, 305$	$2 + \sqrt{5}$	4.236068
5	5-metallic	$x^2 - 5x - 1 = 0$	$0, 1, 5, 26, 135, 701$	$(5 + \sqrt{29})/2$	5.192582

Values are rounded to six decimal places; names beyond bronze vary. Also $\mu_\lambda = \lambda + 1/\mu_\lambda = [\overline{\lambda}]$: its continued fraction repeats the entry λ.

3 / USE BOTH ROOTS TO FIND ANY TERM DIRECTLY

Binet forms: the difference of two powers

Each root power obeys the recurrence: substitute $U_k = r^k$. Matching $U_0 = 0$ and $U_1 = 1$ gives

$$\boxed{U_k = \frac{\mu_\lambda^k - \nu_\lambda^k}{\mu_\lambda - \nu_\lambda} = \frac{\mu_\lambda^k - \nu_\lambda^k}{\sqrt{\lambda^2 + 4}}}.$$

For Fibonacci F_k and Pell P_k, put $\phi = \mu_1$:

$$F_k = \frac{\phi^k - (-\phi^{-1})^k}{\sqrt{5}},$$

$$P_k = \frac{(1 + \sqrt{2})^k - (1 - \sqrt{2})^k}{2\sqrt{2}}.$$

Since $|\nu_\lambda| < 1$, its powers die away, explaining why the successive-term ratios approach μ_λ.

4 / CONSECUTIVE TERMS SATISFY AN INTEGER EQUATION

Pell is hidden in the same quadratic

Put $p = U_{k+1}$, $q = U_k$, and $F(p, q) = p^2 - \lambda pq - q^2$. The next pair is $(\lambda p + q, p)$, and substitution gives

$$F(\lambda p + q, p) = -F(p, q).$$

Starting at $(U_1, U_0) = (1, 0)$, where $F = 1$, proves

$$p^2 - \lambda pq - q^2 = (-1)^k.$$

Complete the square, writing $\Delta = \lambda^2 + 4$:

$$\boxed{(2p - \lambda q)^2 - \Delta q^2 = 4(-1)^k}.$$

This is a **Pell-type equation** for every metallic sequence. In the silver case $\lambda = 2$, division by 4 gives

$$\boxed{(p - q)^2 - 2q^2 = (-1)^k}.$$

Even k gives $+1$; odd k gives -1.

5 / HIGHLIGHT PELL: THE SILVER RATIO MAKES SQUARES TRIANGULAR

Complete the square, then use even powers of $1 + \sqrt{2}$

A number is both triangular and square when

$$\frac{n(n+1)}{2} = s^2 \iff (2n+1)^2 - 8s^2 = 1 \iff X^2 - 2Y^2 = 1, \qquad X = 2n + 1,\ Y = 2s.$$

The silver ratio has conjugate product $(1 + \sqrt{2})(1 - \sqrt{2}) = -1$. Its even powers therefore give solutions with product $+1$:

$$(1 + \sqrt{2})^{2j} = (3 + 2\sqrt{2})^j = X_j + Y_j\sqrt{2}, \qquad n = \frac{X_j - 1}{2}, \quad s = \frac{Y_j}{2}.$$

j	Pell pair (X_j, Y_j)	n	s	$T_n = s^2$
1	$(3, 2)$	1	1	1
2	$(17, 12)$	8	6	36
3	$(99, 70)$	49	35	1225
4	$(577, 408)$	288	204	41 616

Two linked limits, for $\lambda = 2$. $\frac{U_{2j+1}}{U_{2j}} = 1 + \frac{X_j}{Y_j}$: the sequence ratios approach $1 + \sqrt{2}$, and the Pell quotients approach $\sqrt{2}$.

Further reading. J. B. Gil and A. Worley, *Generalized Metallic Means*, §1; K. Conrad, *Pell's Equation, I*, §3–4. Companions: square triangular numbers, continued fractions, and the Pell annulus sieve.

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